Data Challenge 2024 Report: Global Analysis of EV Adoption and Energy Production

Introduction

Electric vehicles (EVs) are transforming the transportation industry, emerging as a cornerstone in the global transition to sustainable energy. As countries commit to reducing carbon emissions, EVs are at the forefront of the shift to cleaner, greener technologies. Nations leading the charge in EV adoption stand to gain significant economic and geopolitical influence, making the EV market a competitive space with strategic implications for the future.

The challenge focuses on three key questions: **1)** How are energy production trends related to EV adoption, and what does this reveal about global attitudes towards green technology? **2)** What factors are the strongest predictors of future EV adoption? **3)** Which countries are most prepared to transition to EVs? By addressing these questions, we seek to uncover the broader environmental, technological, and policy-related drivers of the EV market.

Data Setup

This analysis draws on three key datasets: EV Data, EV Charging Stations, and Energy Production. The datasets were joined by country and year and transformed into a wide format to streamline analysis. Missing values were addressed through interpolation and regression-based imputation where necessary. Given research indicating that temperature impacts EV performance, annual temperature data by country was also incorporated. This combined dataset allows us to explore the environmental, infrastructural, and energy production factors that influence EV adoption trends.

Analysis Part 1: Energy Trends and Their Influence on EV Adoption

First, we looked at how different EV sales correlate with different types of energy (renewable vs. non-renewable). Line graphs over time revealed that positive associations between EV sales and electricity production, regardless of energy type. At a more micro level, BEVs and PHEVs also showed strong correlations with energy types. This suggests that increased energy usage in general has supported EV growth and not exclusively renewable energy. We then looked at individual regions, which told a different story. In most regions, share of renewables and non-renewables are actually converging. In fact, in South America, they are already prioritizing renewables over non-renewables. It turned out that EV adoption is actually related to the increase of renewable energy. The ever-increasing non-renewables share seems to mainly be contributed by Asian and NA countries.

To solidify this observation, we explored annual trends in renewable-to-non-renewable ratio against EV sales using a heatmap. We saw that there is an increase in correlation over the years. Moreover, our t-test revealed a statistically significant difference in renewable-to-non-renewable ratios between countries with high vs. low EV sales. These findings underscore that, while EV adoption aligns with overall energy growth, the upward trajectory of renewable-to-non-renewable ratio implies that countries are increasing their reliance on green technology.

Analysis Part 2: Best Predictors for EV Adoption and Forefront Countries

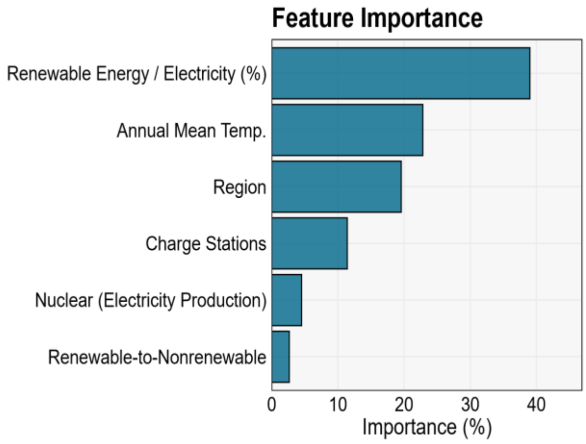
We believe this is best answered with a regression analysis. We employed a 70/30 split to obtain our train and test sets. To avoid large-scale features from dominating the small-scale ones, we applied a standard scaling to all the numerical variables. Setting the response variable to be the log of total EV sales share, we built and compared **two** models:

1) **Linear regression** with LASSO for feature selection and bi-directional stepwise selection. These methods optimize our model and help prevent multicollinearity.
2) **5-fold cross-validated stacking** with LASSO for feature selection. We used kNN, an RBF-kernel Support Vector, and Ridge as base models for our GLM meta-model.

To evaluate the goodness-of-fit, we looked at their MSE, R^2, and MAE on the test data.

Model 1 (Linear Regression)			Model 2 (Stacking)		
MSE	R^2	MAE	MSE	R^2	MAE
2.91	0.42	1.47	2.11	0.58	1.10

Our **stacking model** clearly has better goodness-of-fit scores; thus, we proceeded with identifying the best predictors leading to EV adoption based on Model 2. By plotting a graph of feature importance, we gathered that the 3 best predictors happened to be the **percentage of renewables for electricity production**, **annual mean temperature**, and **region**. We extended this analysis further by uncovering the relationship between the top 6 features and EV adoption. For this, we utilized Partial Dependence Plots (PDPs).



Best Factors that contribute to EV adoption

Positive Influence	Higher renewables share, lower annual temperature, country situated in Europe, more charging stations, higher renewable-to-nonrenewable ratio
Negative Influence	Lower renewables share, higher annual temperature, country situated in Asia / NA / SA, lesser electricity production by nuclear, lower renewable-to-nonrenewable ratio

Now that we have established a solid understanding of the predictors that influence EV adoption, we conducted two methods to unearth the countries at the forefront of this green initiative: 1) **scoring** and 2) **k-means clustering** with 3 centers. Satisfyingly, both methods yield the same result. These countries are **Iceland**, **Norway**, **Canada**, **Austria**, **New Zealand**, **Sweden**, **Switzerland**, **Denmark**, **China**, and **Finland**. Seeing many EU countries make up this list is not surprising as they have all pledged towards becoming a climate-neutral continent by 2050[1]. Led by the Nordics, EVs are just one of the many ways to realize this green initiative.

Conclusion

In summary, our analysis revealed that while EV adoption aligns with overall energy production, a shift toward renewable energy is more prominent in countries leading the EV transition. Factors like renewable energy share, regional location, and temperature emerged as strong predictors of EV adoption, with European nations and regions with robust EV infrastructure leading the way. Our predictive model identified Iceland, Norway, Canada, and several other countries as frontrunners in this green initiative. Some areas of our analysis could be further improved, such as gathering more data to ensure a more comprehensive and externally valid model. Nonetheless, we hope that this report provides a useful framework for understanding the dynamics of EV adoption and guides future efforts in sustainable transportation and energy policy.

1. European Commission. (2022). *2050 long-term strategy*.
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